# TKG Test ROM Manual

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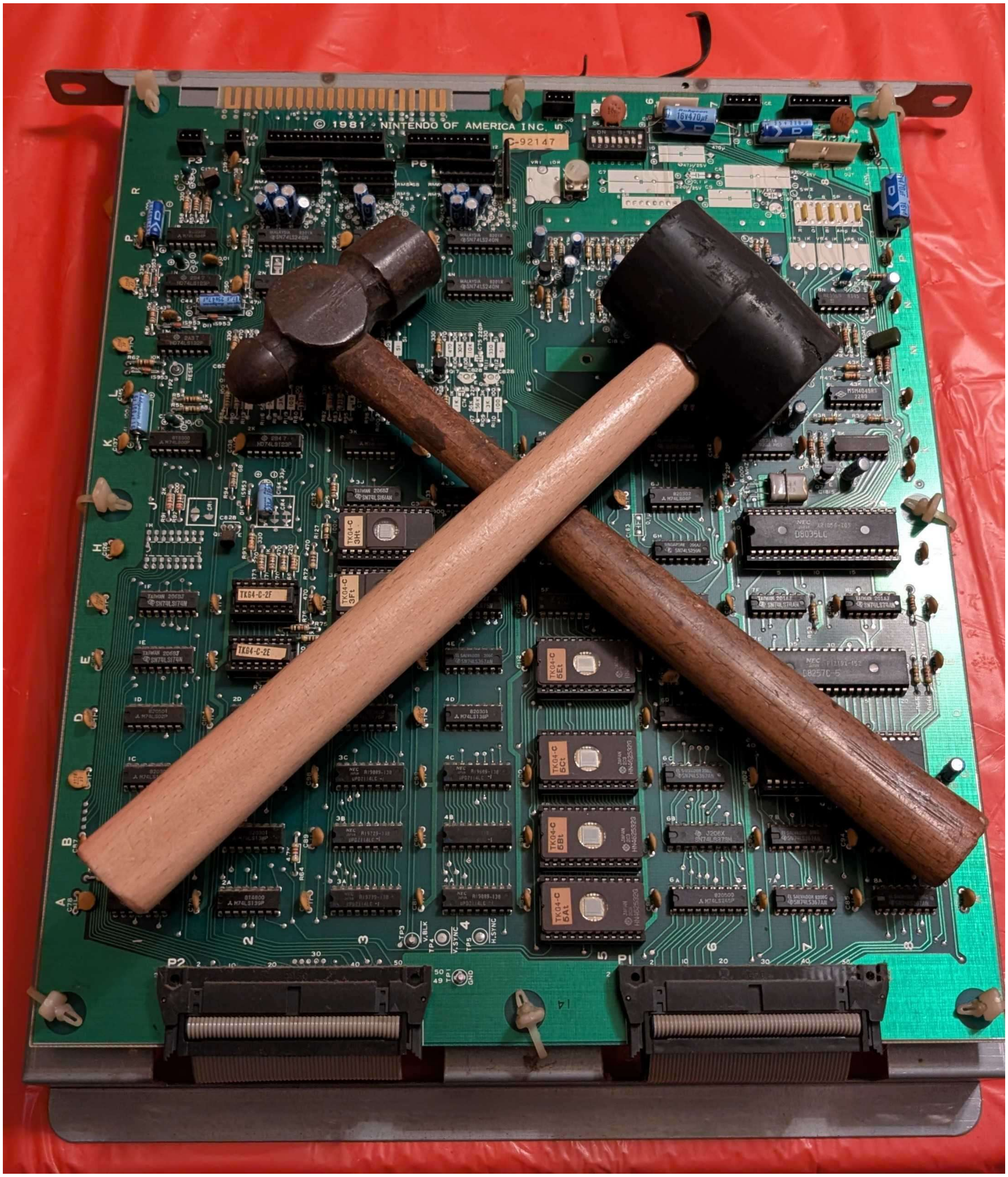


# Table of Contents

# Chapter 1: Overview

## Requirements

This test ROM will not work under every condition. Some things need to be taken into consideration:

- All boards involved must have power, in this case +5V, +12V, and -5V must all be present and within appropriate tolerance, usually within +/- 5%.
- Clocks must be present enough to run. The CPU runs at approximately 3MHz.
- The reset pin and NMI pins must not be held in active. If both are always active, then the ROM will not boot.
- The CPU must be able to access address 0x0000 to 0x0FFF. If the game does not boot it cannot access these ROMs.
- Address and data buffers must be working, if they are not it will not be possible for the program to run.
- Audio OR video must be working. If neither are working, the test ROM will have no way to report failures
- The ROM socket for 5F or 5E (depending on which hardware being tested) must be in good enough condition to make proper contact.

## Compatible Hardware

This test ROM is intended for original TKG hardware. It is compatible with the following hardware:

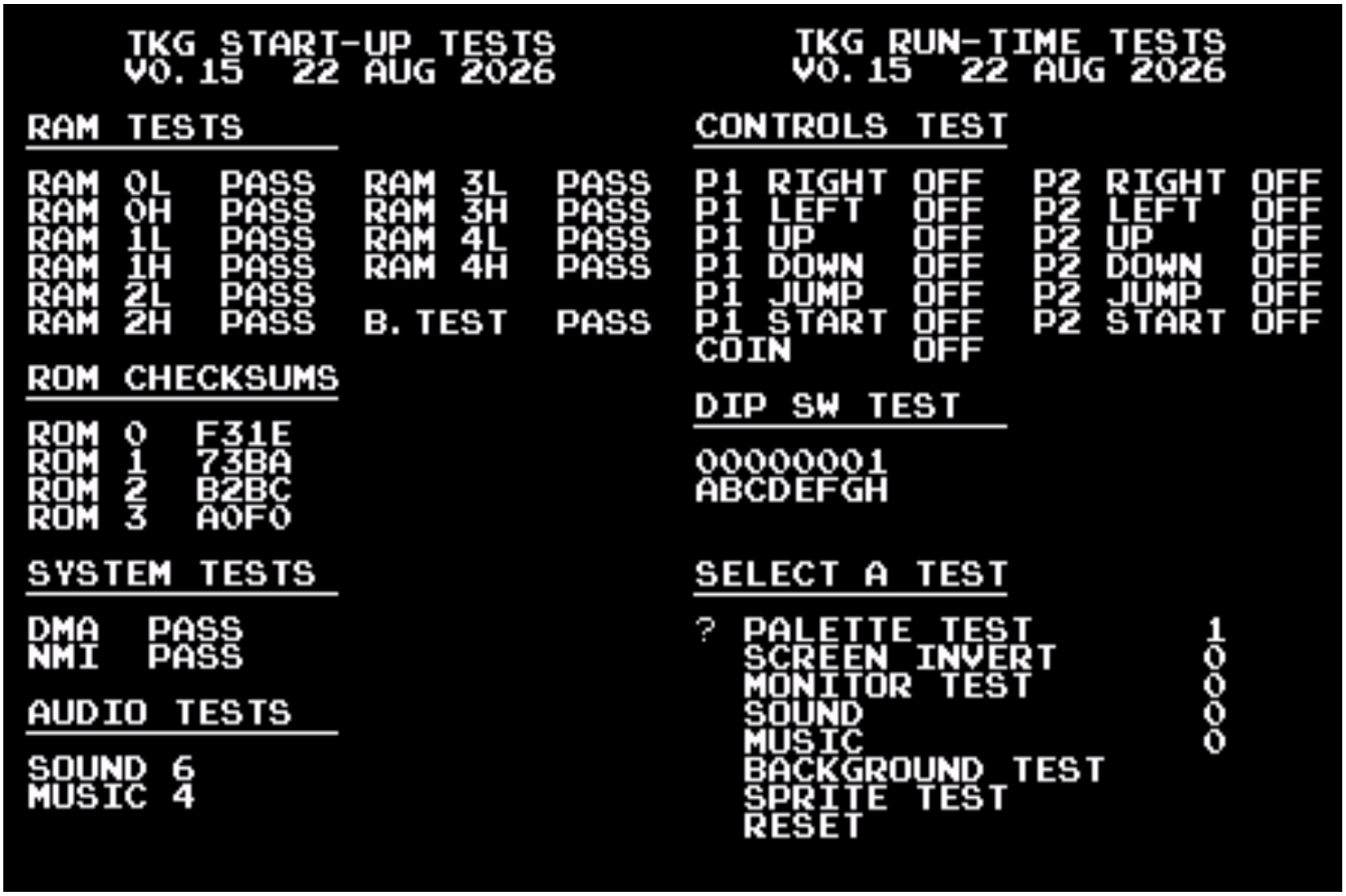
- Fully compatible
  - TKG2
  - TKG3
  - TKG4
- Partially compatible
  - TRS2 – RAM, ROM, DMA, and NMI tests should all work. However, it is a different game so audio tests will not work. Monitor tests might also not work correctly.

This test ROM will work in the cocktail, cabaret, and upright models of the machine.

## Using the Test ROM

Using the test ROM is very simple. The ROM is burnt onto a 2532 4k EPROM. In order to run the tests you place the EPROM into location 5F for TKG2/TKG3 hardware and 5E TKG4 hardware. Note the notch in the socket, EPROM, and pin 1 markers. After the ROM is inserted reconnect the board back to the cabinet or bench setup for testing.

There are two groups of tests performed by the test ROM: Startup and Runtime. Startup tests are tests that are automatically run when the game boots. This tests RAM, ROM, DMA, NMI, and Audio tests. All tests with the exception of audio tests must pass in order to transition from startup to runtime. Runtime tests are additional tests used for diagnosing other issues with the machine in question. It contains things such as a monitor test and control printouts.



# Chapter 2: Startup Tests

## RAM Tests

There are two different sections of RAM tests: RAM test and RAM bank test. The RAM test tests the individual RAM chips themselves. The RAM bank test (labeled B. Test) which tests the addressing of the RAM test. There are five total banks of RAM ICs. Each bank consists of a pair of two 4-bit RAM chips designated high and low where high is the more significant nibble and low is the less significant nibble. The RAM tests are the first tests that run on the system as they need to be run without using RAM for itself.

*Please note: When the RAM test is running you may say "garbage" on the screen. The reason for this is that video and sprite RAM is tested. Whenever data is written to those areas it will appear on the screen automatically. There is no video blanking/supression that can be controlled via software.*

### RAM IC Tests

### RAM Bank Test

### TRS2 and TKG2 RAM Locations

| Name | Designation | Type | Start Address | Size | High Bit Location | Low Bit Location |
|---|---|---|---|---|---|---|
| RAM0 | Work | 2114 | 0x6000 | 1k x 4 | C-6L | C-6H |
| RAM1 | Work | 2114 | 0x6400 | 1k x 4 | C-6M | C-6J |
| RAM2 | Work | 2114 | 0x6800 | 1k x 4 | C-6N | C-6K |
| RAM3 | Sprite | 2148 | 0x7000 | 1k x 4 | V-5J | V-5K |
| RAM4 | Video/Background | 2114 | 0x7400 | 1k x 4 | V-3K | V-3J |

### TKG3 RAM Locations

| Name | Designation | Type | Start Address | Size | High Bit Location | Low Bit Location |
|---|---|---|---|---|---|---|
| RAM0 | Work | 2114 | 0x6000 | 1k x 4 | C-6L | C-6H |
| RAM1 | Work | 2114 | 0x6400 | 1k x 4 | C-6M | C-6J |
| RAM2 | Work | 2114 | 0x6800 | 1k x 4 | C-6N | C-6K |
| RAM3 | Sprite | 2148 | 0x7000 | 1k x 4 | V-2L | V-2M |
| RAM4 | Video/Background | 2114 | 0x7400 | 1k x 4 | V-5L | V-5M |

### TKG4 RAM Locations

| Name | Designation | Type | Start Address | Size | High Bit Location | Low Bit Location |
|---|---|---|---|---|---|---|
| RAM0 | Work | 2114 | 0x6000 | 1k x 4 | C-4C | C-3C |
| RAM1 | Work | 2114 | 0x6400 | 1k x 4 | C-4B | C-3B |
| RAM2 | Work | 2114 | 0x6800 | 1k x 4 | C-4A | C-3A |

| Name | Designation | Type | Start Address | Size | High Bit Location | Low Bit Location |
|---|---|---|---|---|---|---|
| RAM3 | Sprite | 2148 | 0x7000 | 1k x 4 | V-6R | V-6P |
| RAM4 | Video/Background | 2114 | 0x7400 | 1k x 4 | V-2R | V-2P |

## ROM Checksums

The ROM test performs a 16-bit checksum on the contents of all ROMs. This test does not report the integrity of the ROMs nor does it validate that it is correct for a location. That is up to the user to determine. The exception to this is ROM 0, which is where the test ROM is located.

### Test ROM Integrity Test

As part of the ROM test, ROM 0 will be validated. ROM 0 is the test ROM itself. Its own checksum is appended to the footer as part of a post-build script. It will compare the measured checksum with what is expected. If there is any corruption of the test ROM or if there is issues with accessing say the upper addresses, it will report a failure by printing “Integrity Fail”. In this case, the test ROM may not be trustworthy and could run into issues during execution.


### ROM Locations

| ROM ID | ROM Type | ROM Size | Start Address | TKG2/TKG3 Location | TKG4 Location |
|---|---|---|---|---|---|
| ROM 0 | 2532 | 4k | 0x0000 | 5F | 5E |
| ROM 1 | 2532 | 4k | 0x1000 | 5G | 5C |
| ROM 2 | 2532 | 4k | 0x2000 | 5H | 5B |
| ROM 3 | 2532 | 4k | 0x3000 | 5K | 5A |

### ROM Checksum Table

| ROM ID | US Set 1 | US Set 1 Hard | US Set 2 | Japan Set 1 | Japan Set 2 | Japan Set 3 |
|---|---|---|---|---|---|---|
| ROM 0 | A967 | AA53 | A967 | A967 | A967 | A967 |

| | | | | | | |
|---|---|---|---|---|---|---|
| ROM 1 | 73BA | 73BA | 73BA | 745D | 73BB | 745D |
| ROM 2 | B2BC | B2BC | B3AE | B3A9 | B3AE | B3AE |
| ROM 3 | A0F0 | A205 | A384 | A490 | A490 | A4AC |

## DMA Test

TKG hardware relies heavily on DMA to transfer data from work RAM to sprite RAM. During gameplay it is triggered during the NMI routine. The reason for this is because this is when vertical blanking (also known as VBlank) is occurring, meaning that nothing is being drawn on screen. VBlank occurs sixty times per second. On TKG hardware, an 8257 DMA controller is used to perform these operations as well as some supporting circuitry. On TKG4 hardware, that circuitry includes the 74LS373 octal latch at 6B, the 74ls08 quad and-gate at 5F, and the 74LS125 buffer at 6E. The 8257 derives its clock from the 1H signal (~3MHz). Unlike the CPU clock however, it goes through an inverter (74LS04 at 6D). Chip select circuitry is also involved. Refer to the schematics to find out more.

The DMA test works by initiating a transfer from the ranges $6900-$6A80 to $7000-$7180. This test mimics what would be observed in real game conditions. This test is run three times with different byte values. After the DMA transfer is kicked off, the CPU reads back the data that should have been copied over and compares it to what is in the work RAM mirror. If all bytes do not match, then the test is marked as “fail” The DMA test can only occur if RAM2 and RAM3 are functional. If either of these RAM banks failed its RAM test, the DMA test will not process the results and instead print N.A.

A severely failing 8257 DMA controller can cause the board to be unable to boot in rare cases. The DMA controller has direct access to the databus shared with the CPU. For the purposes of the test ROM, you should be able to remove the DMA controller to be able to perform startup tests. However, it will not proceed past the startup tests due to the DMA failure. Be aware that the DMA controller is not always in an IC socket.

## NMI Test

TKG hardware uses a hardware controlled NMI. NMI is derived from the VBlank signal which pulses at approximately 60Hz. It goes through a 74ls74 flip flop (8F on TKG4 hardware) before ultimately toggling the NMI pin on the CPU. The flip flop acts as an NMI enable. To enable NMI, a non-zero value in the lower 4 bits must be written to $7D84. To disable the interrupt, write zero to $7D84. Every time NMI triggers it must it must be re-enabled. In original TKG software, it gets re-enabled towards the end of the NMI routine.

The test ROM validates that the NMI functionality works. NMI is enabled and when it jumps to the NMI routine it sets a bit in the results register. Immediately after enabling NMI, it enters a wait loop. If the NMI routine does not trigger during wait loop it “times out” and from there it will register it as a failure. It is required for the NMI test to pass in order to progress to the runtime tests.

## Audio Tests

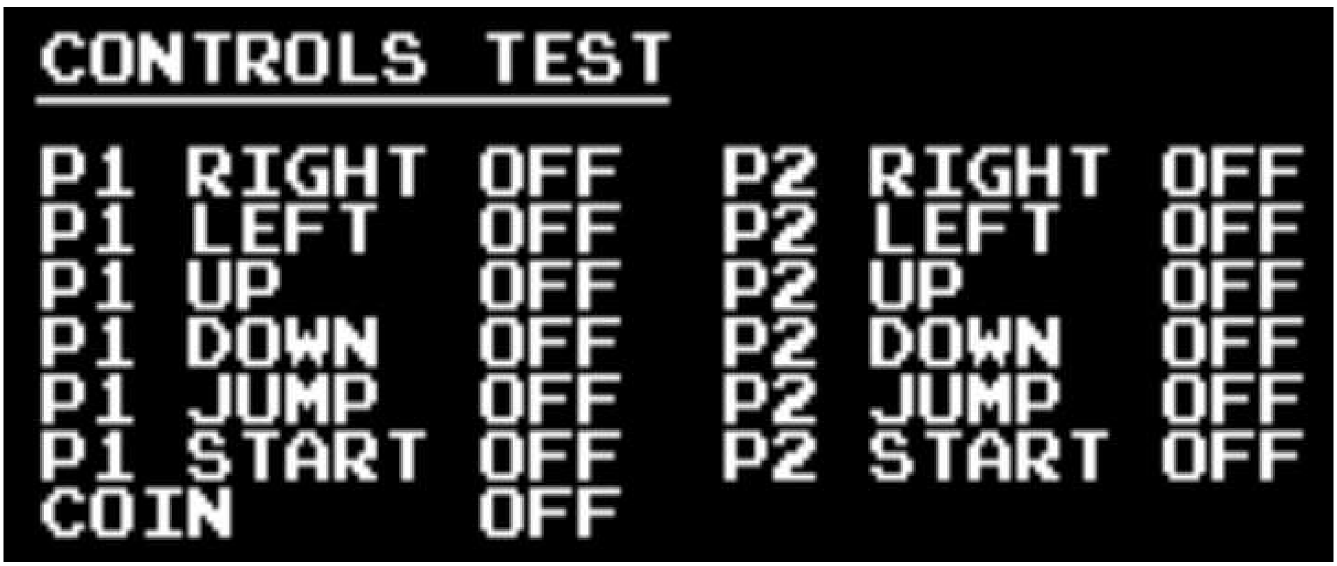
The audio tests automatically play all available sounds on standard TKG hardware. These do not have any pass/fail criteria. It is up to the operator to determine whether or not it is correct. Refer to the tables in the sound and music sections of the runtime chapter to understand what is supposed to be played. There are 7 sounds (0 to 6) and 15 music tracks (1 to 15).


## Transition to Runtime

# Chapter 3: Runtime Tests

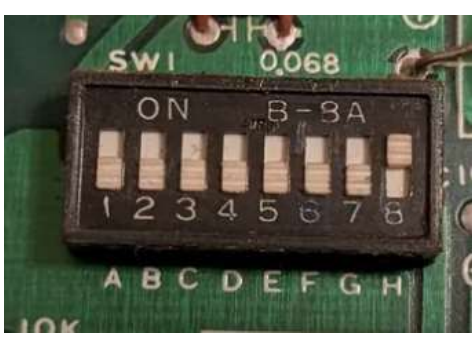
## Controls Test

This is a very simple test that just prints the state of the buttons. This does not have its own screen, it just prints the button state as part of the menu. P1 controls are all the controls on the upright and all available controls on player 1 side of the cocktail table. P2 controls are only available on the cocktail table P2 side. The coin switch is a bit unique. TKG hardware has no way to differentiate between the coin switch and the service switch using software. So if either switch is toggled it will register as “coin”. When a button is pushed, it will be represented by value of the button transitioning to “on”. If any of the values are “stuck” then the first place to check would be the pullup resistor arrays and the buffer ICs. There could also be some failures in the addressing circuitry that is enabling the read of the switches.




## DIP Switch Test

The DIP switches are a set of eight switches found on the CPU PCB. They are labeled both ‘1’ through ‘8’ and ‘A’ through ‘H’. The test will print the current state of the dip switches. “On ”is represented by a ‘1’. “Off” is represented by a ‘0’. To find out what the actual dip switches represent, refer to the game manual. If they are stuck on, like the controls, the first place to check would be the pullup resistors and buffer IC for the dip switches. For more details refer to the schematics on the CPU board.




## Palette Test

This is a very simple test that changes the color palette of the background characters. This will be noticable by the text changing color. The full color palette can be observed using monitor test pattern two. There are four possible color palettes for the background. They can be changed in software by writing to to the least significant bit of addresses $7D86 and $7D87.

## Screen Invert

This is yet another very simple test that just changes the screen inversion status, called flip in the schematics. By default, TKG prints to the screen “upside down”. The actual inversion process is handled through hardware and is enabled by software. To uninvert the screen, write ‘1’ to address $7D82. To invert the screen, write ‘0’ to $7D82. On TKG4, it is controlled by the 74LS259 at 5H and goes through an inverter at 5J. From there the flip signal is consumed by several chips on the video board to handle the flipping of individual video sections.


Normal Screen

Inverted Screen

## Monitor Test

## Sound Test

## Music Test

## Background Test

## Sprite Test

## Reset

This is not really a test. It's only purpose is to enter a loop to wait for the watchdog to reset the hardware. If there is no watchdog present, then it will timeout and jump back to $0000 where all the startup tests will run again. During this waiting period between initiating the reset and the actual reset it will print "reset in progress".

# Chapter 4: Test Architectures

# Abbreviations

**CPU:** Central Processing Unit

**DIP:** Dual Inline Package

**EPROM:** Erasable Programable Read Only Memory

**IC:** Integrated Circuit

**N.A.:** Not Applicable

**NMI:** Non-Maskable Interrupt

**PCB:** Printed Circuit Board

**RAM:** Random Access Memory

**ROM:** Read Only Memory